

SIMATS ENGINEERING

Department of Computer Science and Engineering

CSA15 Cloud Computing and Big Data Analytics

CO2 AT3 - Capstone Cloud Infrastructure Case Analysis and Defense

Editable student template for application-based cloud virtualization and SDDC/cloud operations defense



Assessment

Course Code	CSA15
Course Title	Cloud Computing and Big Data Analytics
Assessment	CO2 AT3 - Capstone Cloud Infrastructure Case Analysis and Defense
Submission Type	Individual digital case analysis and infrastructure defense based on the same capstone title used in CO1, CO2 AT1 and CO2 AT2
Faculty	Dr C. Rajesh Babu
Employee ID	SSETSCS415
Submission Mode	Completed DOCX template and GitHub repository link through Google Classroom

How This Template

Works

- Use the same capstone title used in CO1, CO2 AT1 and CO2 AT2.
- Do not recalculate all VM sizing or repeat all configuration screenshots. Summarize the final AT1/AT2 evidence and use it for design defense.
- Answer all five questions with application-oriented reasoning connected to the capstone product.
- Include sustainable cloud operation decisions wherever relevant: right-sized resources, log retention, storage lifecycle, scheduled shutdown, API-call reduction and VM/container/serverless comparison.
- Paste the final topology diagram in the given response area. The sample diagram is only a structure; redraw it for the student's own product.
- Upload this completed document and supporting evidence to GitHub, then submit the repository link through Google

Assessment Focus

AT1 answered what VM is required. AT2 showed configuration or simulation evidence. AT3 asks whether the student can defend the complete infrastructure design under application, security, operations and failure conditions.

Student and Capstone

Student Name	Shakthi M
Register Number	192511197
Class / Slot	Slot C

Capstone Title	Queueless Canteen: Aware Ordering and Pickup
CO1 Evidence Link	
CO2 AT1 VM Recommendation	vCPU: RAM: GB Storage: GB VM Class:
CO2 AT2 Evidence Link	
GitHub Repository Link	
Submission Date	

Evidence Carry-Forward

Evidence Source	Student Entry	How It Supports AT3 Defense
CO1 concept map / presentation		
CO2 AT1 sizing workbook		
VM calibration sheet		
CO2 AT2 practical evidence		
Capstone architecture decision		

Q1: Virtualization Value and Capstone Cloud Deployment Summary

Explain your capstone product, expected workload, cloud need and deployment context. Analyze how virtualization improves cost efficiency, resource utilization, workload isolation, infrastructure flexibility and cloud deployment readiness for your product.

Design Point	Student Response
Capstone product and users	Queueless Canteen is a cloud-based food ordering and pickup system for students, faculty and canteen staff.
Expected workload	The system will handle menu requests, user logins, food orders, payment requests, order-status updates and pickup notifications. Workload will increase during lunch and break periods.
Cloud need	Cloud infrastructure provides scalable computing, database storage, reliable access and centralized application hosting
Virtualization value	Virtual machines provide an isolated environment for hosting the application, backend and supporting services.
Cost and utilization benefit	Right-sized VMs avoid unnecessary resource usage and reduce infrastructure cost. Resources can be increased when demand grows.
Isolation and flexibility benefit	Application services can run in isolated environments, reducing interference between services and making maintenance easier.
Case Analysis Response: Write 8-12 technical lines connecting virtualization value to your capstone product. Queueless Canteen is a cloud-based application designed to reduce physical queues in a college canteen. Students can view the menu, place orders and receive pickup information through the application. The workload is normally moderate but can increase significantly during lunch and break periods. Cloud deployment allows the application to remain accessible to many users simultaneously. Virtualization allows the application to run on a dedicated virtual machine with controlled CPU, memory and storage resources. The VM can be right-sized according to the expected workload to avoid unnecessary resource consumption. Virtualization improves resource utilization because physical infrastructure can host multiple isolated virtual workloads. Application and database services can be separated to improve reliability and maintenance. VM-based deployment also allows backup, snapshot and recovery operations. Cloud resources can be scaled when the number of users or orders increases. Therefore, virtualization provides flexibility, isolation, cost control and cloud deployment readiness for Queueless Canteen.	

Q2: Rapid Provisioning, Testing and Virtual Machine Lifecycle Case

Using your CO2 AT1 VM recommendation and CO2 AT2 practical evidence, explain how virtual machines, VM images, templates, snapshots, clones or containers support rapid provisioning, testing, deployment, rollback and maintenance for your capstone product.

Lifecycle Element	AT1/AT2 Evidence Used	Capstone Use / Justification
VM image or base environment	Ubuntu/Linux server with required runtime	Provides a consistent environment for the application.
Template or repeatable setup	Standard application configuration	Allows another VM to be created with similar settings
Snapshot / rollback	VM snapshot before major updates	Allows recovery if an update causes application failure.
Clone / testing copy	Clone of development VM	Used to test new features without affecting the main application.
Container or VM decision	VM selected for prototype	VM provides simple and easy-to-understand infrastructure for the project.
Maintenance and update plan	VM selected for prototype	Updates are tested first and then deployed to the main environment.
Case Analysis Response: Explain how lifecycle practices reduce deployment delay and risk. VM lifecycle practices make the Queueless Canteen deployment easier to manage and reduce deployment risks. A base VM image can contain the operating system, application runtime and required dependencies. A repeatable VM configuration helps create a similar environment whenever required. Before major updates, a snapshot can be created so the previous working state can be restored if a problem occurs. A cloned VM can be used for testing new features and application updates. Containers can be considered later if the system is divided into multiple independent services. Regular maintenance, security updates and backups help keep the system reliable. These practices reduce deployment time and prevent testing activities from affecting the live canteen application.		

SOA/API Integration Explanation: Explain how services communicate and why modular service design is useful.

The Queueless Canteen application follows a modular service-oriented architecture in which different application functions communicate through APIs. The frontend communicates with the backend using secure HTTPS REST APIs. The backend handles authentication, menu management, order processing and pickup-status management. The database stores users, menu items, orders, payments and order-status information. A notification service can send order-ready messages to users. Analytics can process order and crowd information to identify busy periods and improve preparation planning. This modular approach allows individual services to be updated without redesigning the complete application. API-based communication also makes it easier to integrate future mobile applications, payment systems and analytics services.

Q4: Design Alternatives, Identity, Access and Federated Cloud Privacy Case

Compare at least two deployment alternatives for your capstone product, such as single VM, multi-tier VM, containerized deployment, managed backend, federated cloud or hybrid cloud model. Justify the selected design by analyzing identity management, role-based access control, federated access, data sharing, privacy protection and security risks.

Deployment Alternative	Advantages	Limitations / Risks	Identity and Privacy Impact
Alternative 1	Simple, low cost and easy to configure	Single point of failure and limited scalability	Centralized authentication but weaker isolation
Alternative 2	Better separation, scalability and maintenance	More complex to configure	Better access control and service isolation
Selected design	Flexible, scalable and easier to maintain	Requires proper configuration and monitoring	Provides stronger separation of users and services
Security Area		Student Decision	
User roles		Student, Faculty/Staff, Canteen Staff and Administrator	
Authentication method		Username/password with secure session or token-based authentication	
Role-based access control		Each role receives only the permissions required for its activities	
Federated access / partner access		Optional college identity/SSO can be integrated in future	
Sensitive data handled		User information, order information and payment-related transaction data	
Privacy protection decision		Encrypt communication using HTTPS, restrict database access and store only required information	

Case Analysis Response: Defend why the selected design is better for identity, access and privacy.

The selected multi-tier cloud design is better than a single VM because it separates the frontend, backend and database responsibilities. Students should only be allowed to view menus, place orders and check their order status. Canteen staff should be allowed to manage menus and update order preparation and pickup status. Administrators can manage users, system settings and reports. Role-Based Access Control prevents users from accessing functions that are not required for their roles. Authentication protects user accounts, while HTTPS protects information during communication. Database access should be restricted to the backend service rather than directly exposing the database to users. Only necessary user and order information should be stored to improve privacy.

Q5: Software-Defined Datacenter and Cloud Operations Defense Case

Analyze how your infrastructure will respond to operational and failure scenarios such as VM failure, database growth, traffic spike, credential leakage, service outage, resource contention or cost increase. Explain how SDDC concepts and cloud operations practices such as automated provisioning, software-defined networking, software-defined storage, monitoring, logging, backup, rollback, scaling, resource control and maintenance improve the final infrastructure design. Include Green Computing decisions such as resource right-sizing, storage lifecycle control, log retention, API-call reduction, scheduled shutdown, VM/container/serverless comparison and carbon/energy proxy indicators where relevant.

Operational / Failure Scenario	Expected Impact	SDDC / Cloud Operations Response	Evidence or Control
VM failure	Application becomes unavailable	Restart/recreate VM using image or backup	VM image, snapshot and monitoring
Database growth	Storage may become insufficient	Increase storage or migrate to scalable managed DB	Storage monitoring
Traffic spike	Slow response during lunch period	Increase resources or add application instances	CPU/RAM monitoring and scaling
Credential leakage	Unauthorized account access	Reset credentials and revoke active sessions/tokens	Authentication logs
Service outage	Users cannot place/check orders	Backup service or restart failed component	Health checks and monitoring
Resource contention	Application performance decreases	Resource limits and right-sizing	Health checks and monitoring
Cost increase	Higher cloud bill	Review resource usage and remove idle resources	Cost monitoring
Operational Readiness Area		Student Plan	
Automated provisioning		Use VM images/templates to quickly recreate application environments.	
Software-defined networking		Use virtual networks, security rules and controlled access between application components.	
Software-defined storage		Use scalable cloud storage and database storage based on demand.	
Monitoring and logging		Monitor CPU, memory, storage, application errors and API requests.	
Backup and rollback		Perform database backups and VM snapshots before major changes.	
Scaling and elasticity		Increase VM resources or application instances during high-demand periods.	
Resource control and maintenance		Use right-sized resources, update software regularly and remove unused resources.	
Green Computing and sustainability control		Reduce idle resources, control storage retention and minimize unnecessary API requests.	
Green Cloud Decision		Student Justification	
VM right-sizing from CO2 AT1		Select CPU, RAM and storage according to actual application requirements rather than over-provisioning	
VM vs container vs serverless choice		Use a VM for the initial prototype because it is simple to configure; containers/serverless can be considered when workload increases	
Scheduled shutdown / idle-resource control		Development and testing VMs can be shut down outside working hours.	
Log and file retention policy		Keep recent operational logs and remove/archive unnecessary old logs.	
Storage lifecycle and backup cleanup		Remove duplicate and obsolete backups while retaining important recovery copies.	
API-call or polling reduction		Use event-based notifications or reasonable polling intervals instead of continuous unnecessary requests.	
Carbon/energy proxy indicator to track		Track VM CPU utilization, uptime, storage usage and number of active resources as operational energy proxies.	

Final Infrastructure Defense Statement: State whether the design is ready for prototype, pilot or production. Defend reliability, security, SDDC/cloud operations and Green Computing improvements required next.

The Queueless Canteen infrastructure is suitable for the **prototype and pilot stage**. The design uses cloud virtualization, API-based communication, centralized database storage, authentication, role-based access control, monitoring and backup mechanisms. The system can handle increased demand during peak canteen hours by scaling application resources. VM snapshots and database backups provide recovery capabilities in case of failures. Security can be improved through HTTPS, strong authentication, RBAC and restricted database access. Green Computing can be improved through VM right-sizing, scheduled shutdown of development resources, storage lifecycle management and reduced unnecessary API calls. Before production deployment, load testing, automated scaling, stronger monitoring, disaster recovery and security testing should be implemented.

CO2 Reflection and Individual Defense

This section must be individually written. It should show what the student understands from the complete CO2 chain: infrastructure sizing, practical configuration evidence and final operations defense.

Reflection Prompt	Student Response
What is the strongest infrastructure decision in your capstone design?	The strongest infrastructure decision is using a cloud-based multi-tier architecture with separate frontend, backend/API and database components. This improves scalability, security, maintenance and resource management.
Which CO2 concept became clearer after AT1, AT2 and AT3?	The concept of virtualization and cloud infrastructure lifecycle became clearer. AT1 helped in selecting the required VM resources, AT2 provided practical configuration experience, and AT3 helped understand how the infrastructure should handle security, failures, scaling and sustainability.
Which risk is most serious for your capstone: failure, cost, privacy, scaling or sustainability?	Scaling is the most important risk because the number of orders can increase significantly during lunch and break periods. If the infrastructure cannot handle the sudden increase in requests, users may experience slow response or failed orders.
What evidence would you show during viva to defend your infrastructure design?	I would show the VM sizing recommendation, deployment architecture, application screenshots, API integration, database design, authentication and RBAC implementation, monitoring results, backup strategy and the final cloud topology diagram..
What will you improve before final capstone deployment?	I would improve the system by adding automated scaling, stronger authentication, better monitoring, database backup automation, load testing, improved notification handling and stronger privacy controls. I would also optimize cloud resources to reduce unnecessary cost and energy consumption.

Student-Facing Assessment

Criteria	Descriptor
Capstone continuity	The answer uses the same capstone title and connects CO1, CO2 AT1 and CO2 AT2 evidence.
Virtualization and lifecycle reasoning	VM, image, snapshot, clone, container or managed-service decisions are technically justified.
Topology and integration clarity	Architecture diagram and SOA/API explanation show clear module interaction.
Identity, access and privacy	Roles, authentication, RBAC, federated access, sensitive data and privacy decisions are defended.
SDDC/cloud operations	Failure response, monitoring, logging, scaling, backup, rollback and resource control are addressed.
Green Computing defense	Right-sizing, retention, API efficiency, idle-resource control and sustainable cloud choices are explained.
Submission quality	GitHub, template completion, evidence references and individual reflection are complete.

GitHub and Google Classroom

Repository Item	Expected Content
README.md	Capstone title, student details, final VM recommendation, deployment topology summary and AT3 evidence index.
Completed AT3 document	This completed editable DOCX or exported PDF if instructed.
Diagram image	Final topology diagram if created separately.
Evidence references	Links or screenshots from CO1, CO2 AT1 and CO2 AT2 as needed.
Green Computing note	Short README section explaining sustainable cloud choices and resource-control decisions.
Google Classroom	Submit the GitHub repository link through GCR.

Student Assurance

I confirm that this case analysis, design reasoning, topology diagram and final defense statement were prepared individually by me for the stated capstone title. I have not uploaded passwords, tokens, private keys or sensitive account information.

Student Name	
Register Number	
Signature	
Date	

Faculty

Faculty Name	Dr C. Rajesh Babu
Employee ID	SSETSCS415
Constructive Feedback / Remarks	
Strength Observed	
Improvement Suggested	
Signature / Date	